Weston Kirk

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Education

Stanford University *Sept 2022 – December 2027*
BS in Mathematics, Earth Systems
MS in Computer Science

- **Relevant Coursework:** Automated Reasoning, Probability Theory, Linear Algebra, Continuous Math for ML, Discrete Math, Number Theory, Graph Theory, Systems, Remote Sensing, Geosciences
- **Research interests:** Neurosymbolic AI, Discrete Mathematics, AI4Math, AI4Science

Publications

VeriBench: An End-to-End Formal Verification Benchmark for AI Coding Agents in Lean 4. Brando Miranda, Srivatsava Daruru, Ethan S. Hersch, Zhanke Zhou, Allen Nie, Daneshvar Amrollahi, Leni Aniva, Iddah Mlauzi, Kirill Acharya, Elyas Obbad, Dilara Soylu, **Weston Kirk**, et al. ***ICML 2026 Workshops AI4Math and DL4C*** (Poster). Also accepted to the ***2025 ICML AI for Math Workshop***.

Experience

Research Intern *Jul 2026 – Present*
Accio Data *Austin, TX*

CURIS Fellow, STAIR Lab *Apr 2025 – Jun 2026*
Stanford CS Department *Stanford, CA*

- Researched formal verification benchmarks for AI code agents in Lean 4 under Dr. Sanmi Koyejo.

Biology Researcher, Daru Lab *Sept 2023 – Sept 2024*
Stanford Biology Department *Stanford, CA*

- Studied crop wild relatives biogeography under Dr. Barnabas Daru.

Earth Systems Public Service Fellow *June 2023 – Sept 2023*
Haas Center for Public Service, Save Our Springs Alliance *Austin, TX*

- Co-founded a conservation coalition; presented at the inaugural event with 50+ attendees.

Projects

Comparison of Discrete Logarithm Algorithms *2025*

- Compared Baby-step Giant-step, Pollard's Kangaroo, and Pohlig–Hellman algorithms for the Discrete Log Problem, deriving complexity bounds and cryptographic security tradeoffs.

BSURP Crop Wild Relatives Research Project *2024*

- Developed algorithm for n-nearest monophyletically clustered relatives; modeled diversity shifts.

CS 109 Grand Challenge Project *2024*

- Used rolling MLE for Poisson/Exponential RVs to analyze spring-flow time series; finalist (9 of 113, ~8%).

Additional Information

CURIS Fellowship *2025*

- Selected as 1 of 15 undergraduates for CURIS research fellowship.

YC AI Startup School *2025*

- Admitted to inaugural conference (3k/18k ~17%).

Reviewer, NeurIPS 2025 MATH-AI Workshop *2025*

- Selected to serve as a reviewer for a premier workshop on mathematical reasoning and artificial intelligence.

CS 103 Challenge Problem Award *2024*

- Solved advanced theory problems on diagonalization, Turing Machines, and regular languages.